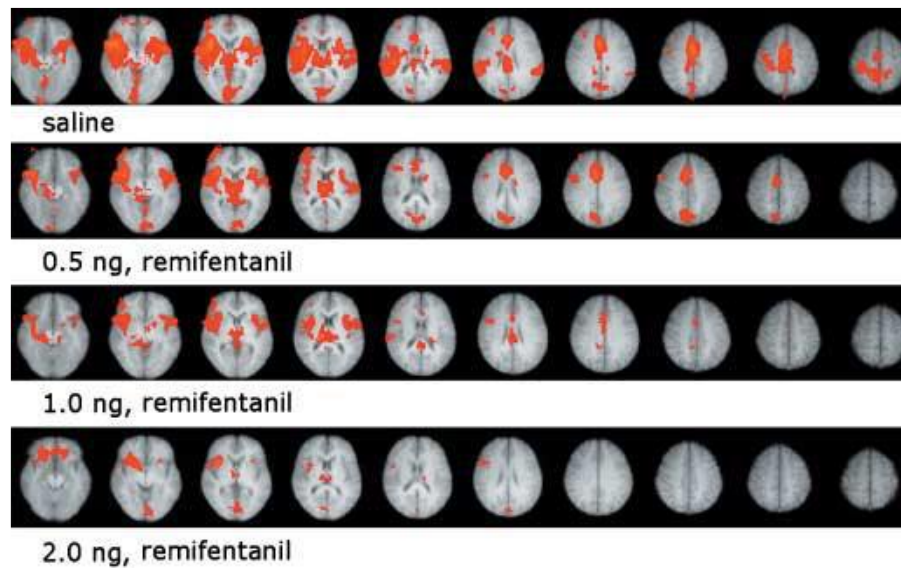


The higher is the concentration of analgesics the lower is the response, primarily in the anterior cingulate cortex. This is what you can study, the effects of opioids. If you give placebo, 30% of people improve.



The interesting part for us is to have a modulation of pain in the anterior cingulate. There were several examples of PET studies measuring blood flow with patients susceptible to achnesia? and through hypnosis they were having them to perceive as less or more unpleasant their handling of cold water. Only saying that they were going to get a painful stimulus, in the anterior cingulate will appear a higher response.

For most of these studies you could claim that the anterior cingulate is not really the pain processing affective component, but sort of like decisional aspect on whether to react to painful stimuli. This portion of the brain is actually getting information directly from the spinal cord, the thalamus, to modulate our behaviour on pain perception.

Vision

Here we would see a much finer degree of description and complexity of representation, as vision is the most well extended studied modality across all our sense.

The information is seen in two dimensions by the two eyes and then processed 3D mentally in our brain. There are at least two parallel and interacting pathways that process information. More than half of the cortex is devoted to visual processing. The whole occipital lobe is subserving visual processing; there is no other region that does the same for a single function.

What we actually can say is that the visual system transforms transient like patterns (even from low level features we can reconstruct the image) on the retina in to stable representation of the tri dimensional world.

The eyes adapt and change their shape to get the light getting properly to the posterior side of the eye. But when the shape of the eyes changes and the ability to modulate this accommodation is lost, we typically have disorders that need correction to improve our ability to see. The eyes have a protective surface that is mostly transparent (in the cornea), and then the rest of the eyes is the sclera and is mostly peripheral to this central transparent part. Then we have two chambers (anterior and posterior) divided by the lens, and these are filled with two aqueous solutions, the

aqueous humor is much more fluid, and the vitreous humor in the back has much more of a gel-like structure. The opening between the two chambers is called the pupil, and it has the specific role of modulating the number of photons that can get to the retina, in a light environment the pupil shrinks.

The retina is the internal layer of the chamber of the eye, and on the posterior part there is a pigmented epithelium, a dark layer full of melanin that works like the black screen of the movie theatre, since it avoids the scattering of photons on retina. These layers are also full of blood vessels that feed the retina, that is made up of three layers of cells, an external layer that is made up of photoreceptors, a second layer that is made up of bipolar cells, and finally a third layer that is made up of ganglion cells. The axons of the ganglion cells are bundled together into what we call the optic nerve, and it is typically entering the back part of the eye and drags the spot without photoreceptors, and we typically call that blind spot (the finger disappears at the centre).

Light photons have to cross all three layers of cells. It appears to be an efficient system, because they reach a region full of blood vessels, they are more fed.

There are also two horizontal layers, the horizontal cells stand between the photoreceptors layer and the bipolar cells; the amacrine cells stand between the bipolar and the ganglion cells. You understand immediately that this number of layers is telling you that there is a lot of processing, yet at this level.

There are photoreceptors that have a one-to-one relationship with bipolar cells, so their information should be very important, but this information also interacts with neighbour photoreceptors, so there should be some population coding at the level of the retina that captures information and starts doing some specific processing per se.

Photoreceptors in the retina do not exhibit action potential, rather light activation causes a changing in membrane potential, and it comports the release of a neurotransmitter. This regulation of the closure of sodium and calcium channels are telling that the darker the situation and the least intense is the depolarization, the higher is the light, more negative is the response.

One important feature is that this biochemical cascade initiated by capturing photons, provides an enormous amplification of the signal. One single molecule of rhodopsin activates 800 molecules of transducin. These molecules initiate a cascade of events that modulates the opening and closing of the channels.

We have two types of photoreceptors, rods and cones, that are not just established by the shape, but by the photopigment they contain. They have different distribution density along the retina, and also a different pattern of connection to the bipolar and the ganglion cells.

- The rod system has a very low spatial resolution, but is extremely sensitive to light, they are primarily distributed at the periphery of the retina and are primarily used when we refer to mesopic and scotopic visual conditions (reduction of light). An example of when you use rods is when you look at stars, you see them with the periphery and not concentrating on them.
- The cone system has a very high spatial resolution, is relatively insensitive to light and are much more sensitive to acuity, they are concentrated in a specific portion of the retina.

Cones have very low presence at the periphery, are all concentrated in a very small portion, corresponding to the visual field, and it is called fovea. Rods are almost absent in the fovea; they have higher representation in the periphery. In the fovea you also have a mechanism that opens

the different layers and create an area called little fovea, where light doesn't have to cross the two internal layers (even blood vessels), but goes directly in the photo receptors (no distortion or scattering).

In terms of pattern transmission, cones (with a higher acuity) typically are projecting to the polar and ganglion cells directly. This information is also mediated with other cones and rods, but this kind of response is much more mediated by more than one cell.

When the stimulation gets to the ganglion cell, whose receptive field gets the light spot, it responds to it with a slowly adapting response but the whole centre is not responding. There are different patterns of response, that allow the retina to discriminate contrast and contours.

The cone release glutamate, but it can be excitatory and inhibitory. The different cones are already allowing us to process information and get it to the brain. The issue is that this kind of processing is done by our retina, our brain has no impact, it just receives information, that could be distorted by disorders at the level of the eye.

Information from retina is sent to the posterior part of the brain through the optic nerve, that merges into what is called the optic chiasm. Information is typically divided into two elements, the nasal part of the retina and the temporal part of the retina.

- The nasal part is primarily providing information on the ipsilateral visual field.
- The temporal part provides conveys information on the contralateral hemi field.

When the two optic nerves merge into the chiasm, the nasal part is still processing. On the left brain you will get information from the left temporal and the right nasal retina, conversely the right hemisphere receives information from the right temporal and the left nasal. After passing in the optic chiasm, there are two portions that are called the optic tracts, that send information to thalamus (primarily the lateral geniculate nucleus), and this information is all related to the opposite hemi field. This information from the thalamus is then projected to the optic radiations (brain fibers) back to the occipital cortex.

The ganglion cells are providing very specific information or gross information; there are ones that only say if the light is on or off. This is something very useful for the hypothalamus, with the regulation of circadian rhythm. There are blind people (born blind) that has no light perception, and their circadian rhythm is all altered.

The muscle that regulates the movement of the pupil is called the pupillary constrictor muscle, and it relies on the changing in light. To see if there is damage at the level of the brain stem, you can get light to the pupil and it has to shrink, if there is no damage this reflex is very simple direct and is a good way to demonstrate that that part of the brain is really working.

The fourth pathway goes to the superior colliculus, a deep structure in the diencephalon, which has a huge importance, because it orients the movement with coordination with the eyes. It doesn't only rely on vision but also with other sensory modalities.

When we have a visual field, we can divide it in horizontal and vertical axis, and we can select four portions of it. If the two axes are defined with a fixation point you have a portion of the visual field that is actually captured by both eyes, but you have also portion of the peripheral visual field that is captured only by the nasal portion of one eye, and this information is sent contralaterally.

This very specific and defined representation is shared with the cortex, and we have a specific representation at the level of the occipital cortex. The right occipital lobe receives information of

the left visual field, with the upper portion that is represented in inferior dent of the calcarine sulcus, and the inferior visual field is represented with the upper dent of the visual cortex. The most central portion of the visual field is represented posteriorly in the occipital pole, while moving anteriorly I'm going from the foveal to the periphery of the brain.

Visual pathway is really complex. Any damage along the path is creating different elements of loss of the visual field, something that is called scotoma, something that is hidden.

V1 (visual cortex) is divided in columns. Columnar organization is primarily referring to functional organization across all six layers of the cortex, and vertically you can see how a single column responds to the same orientation, while if I'm looking to a larger chunk of the cortex we have orientation selective columns. Together with orientational columns, we have (orthogonally) ocular dominants columns, all together we have hyper column.

Information from the retina is getting first to the lateral geniculate nucleus, that is made up of six layers, which are made up by 2-3 families of neurons, 4 layers are made up of parvocellular neurons and these are indeed connected with ganglion cells in the retina that are already smaller, so in the retina we have small ganglion cells that project to the four layers of the lgn. Then we have large ganglion cells (magnia) that project to the layers 1 and 2 of the thalamus, finally in the middle of these layers we have a layer made up of koniocellular neurons.

This p and m ganglion cells are so important because they have different functional features, magnocellular layers in the lgn are not sensitive to colour contrast but have a very high temporal frequency, the p cells have a very high spatial frequency. The information they are offering is actually processed across two different pathways, the parvocellular pathway is processed more ventrally, along what is called the inferior temporal (ventral pathway), that is primarily devoted to process informational identity (colours, shape, texture). The m cells are going to project to regions after V1, devoted to motion, and then to the posterior parietal cortex, this is called the dorsal parietal pathway, that processes much more information about movement and space.

We don't have two pathways that process different pieces of information. The novelty of this is that already from the retina there is a functional organized information that get to the thalamus and then to the brain. The feature of the cells that are present in the retina and then in the thalamus are already constraining and impacting the information that we are able to process from the external world. This information then will be projected and processed to two different pathways, one whose information for us is much more important for movements and action (dynamic), while the ventral pathway is much more related to semantic, language, knowledge organization, related to the identity and the shape of the object.

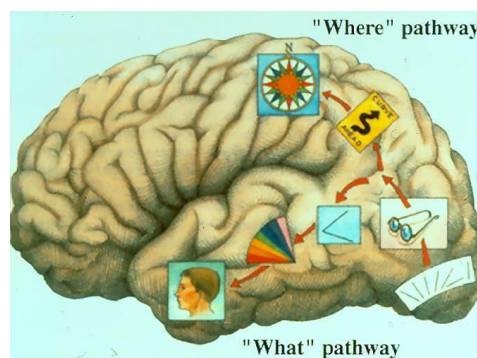
The important aspect is that this kind of functional subdivision is actually starting directly from the retina, the retina itself has sensory component and is already capturing specific features from the visual stimulus, and conveys this to the thalamus (specifically to the lateral geniculate nucleus) and leads backwards to the occipital cortex where, in the striate portion, we have the demographical representation of the visual field.

Subsequently this piece of information follows two different pathways, one moves inferiorly ventrally towards the temporal cortex. These specific pathways appear to convey information primarily on the parvocellular (p ganglion cells) and the magnocellular pathway, that conveys information ventrally towards the parietal cortex.

Besides the two functional aspects and subdivision of the parvocellular ventral stream, that conveys more information with higher spatial resolution, so primarily related to definition and discrimination of shape, texture, orientation and colours. This pathway is actually characterized as a sort of “what” pathway, that is telling me what I’m actually perceiving. This information relative to the consequent recognition of the items I’m perceiving would be therefore related to also understanding a semantic, linguistic interpretation of what I’m seeing. Indeed the information that progressively moves from the early visual areas in the occipital cortex, anteriorly to the temporal lobe, are actually involved in this knowledge organization and semantic processing because these regions are bridging the auditory information with the visual one (the superior-temporal gyrus is not just a region of multisensory integration but is a region involved in speech and language processing). Therefore, we have a functional merging of sensory information (I think in the superior colliculus) coming from the occipital cortex with all the speech component of language.

Every time you have a specific lesion in the anterior region of the temporal lobe, we have a formal neurodegeneration that is more often called semantic dementia.

We have to always keep in mind that a structural and architectural organization of the brain, always mirrors the anatomical features. Similarly, more dynamic information is progressively going dorsally, this piece of information of the magnocellular pathway is much more related to movement perception, dynamical changes of stimuli, and stimuli that change are also changing location in space, and all this piece of information is important to integrate visual information with the information coming from the sensory motor cortex. This pathway is called the “where” or “how” visual pathway and describe and collect all pieces of information that are useful to motor action, or to process objects spatially. That is why when we have a lesion of this pathway, we have the inability to perform actions and motor sequences (ataxia).



These two pathways are strongly bonded in visual perception but appear to process specific content of information independently from the sensory modality, to which this information is conveyed to the brain. This means that if you close your eyes and recognize by touch a bottle you will recruit the parvocellular pathway, so shape is represented in that portion of the cortex if you acquire information with a sensory modality that is nonvisual. Consequently, we also can do motion discrimination by hearing the sound of a car or a bike, or the flow of water in my hands.

It seems like that this fine architecture of the two pathways, nonetheless, looks like it is much more specific to compute and represent some specific piece of information independently of the sensory modality that conveys the information to the brain.

For many years we have been describing these two as the main visual pathways, but we are also thinking that there is a third, going in the middle in the superior temporal cortex, in the supra-

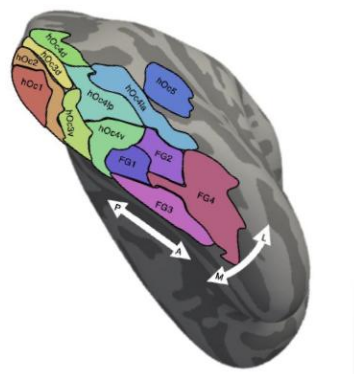
marginal and angular gyri. This third pathway is much more related to social cognition, the interaction with other people.

Vision has been the most studied sensory modality, since it is our most dominant modality. For many years we have been labelling these two pathways as visual associative cortex, it is not only visual. It looks like each portion of the brain computes specific item and piece of information independently of the sensory modality that conveys this information to the brain.

It is impossible to recognize the shape of objects in an auditory way. Furthermore, I'm able to recognize the entity of an object but not the shape, there are pieces of information that could be studied across modalities, like motion.

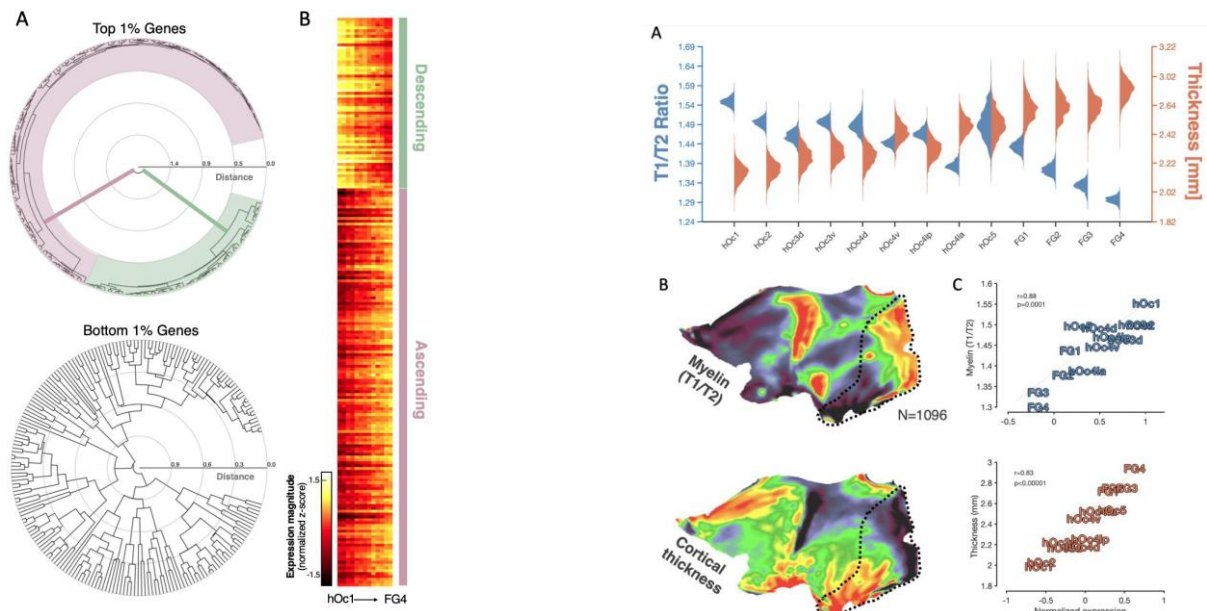
What happens when you close your eyes and you recognize an object by tactile modality? You recall a visual image. This is imagery, and you activate the visual cortex. This activation is actually related to something that is computationally independent from the sensory modality, and it's not related to visual imaging. This is something that happens also in people blind from the birth, when they are asked to recognize an object, they recall the same areas.

In a paper published on top journal, a study relied only on publicly available datasets, they never took a measure. They looked at genes expressed in the tissues of the brain's area in figure and they defined different portion on the cortex (parcellation), and they go from early visual areas to the more anterior temporal regions. If you look at the expression of some genes you can define two gradients: ascending and descending, from the occipital cortex to the temporal cortex.



The whole body relies on these gradients in order to orient the exact position of body parts, organs, and furthermore if there is one body part that should be organized is the brain. This difference of expression in genes is a way to which genes are actually differentiating, for example the occipital from the temporal lobe.

When we look at other measures, for example the amount of myeline in the cortex and the thickness, one thing that we can observe is that the early occipital areas have a high amount of myeline but are very thin, while brain areas on the temporal pole appear to be thicker but have a less amount of myeline.



Then they looked at another dataset, that concentrated in getting different gene expression across different ages and the ages are not just after birth, but also in post-conception weeks. In the newborn brain you can see that the ascending and descending gradients are actually going to develop, then one thing that you observe when the baby is still in the uterus is that the ascending gradient is already quite defined, while the descending gradient is not. The clear maturation of these two gradients is actually occurring around 5 years, so it occurs this amount of time for the visual system to completely form and develop, and between the two gradients the ascending one (related to thickness of the brain) appears to develop earlier, even before birth. What is interesting is that it really takes a lot for the visual system to get a stable organization and maturation, and we should consider this as a sensitive period in which sensory experience is actually impacting the brain.

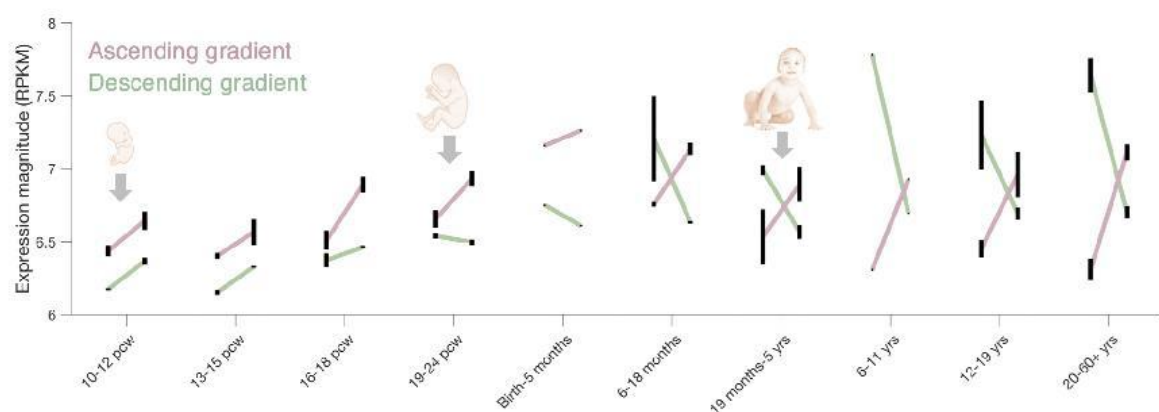


Figure 5: Ascending and descending gene gradients emerge at different points in human development. The two gene gradients

What is quite interesting is that you can compare these genes also in humans e non-human primates, and while the ascending gradient is quite similar, the monkey's visual system has no

descending gradient. So, for sure this descending gradient that is developing during childhood is very important for a more complex and complicated visual representation.

What we get from the analysis of this big dataset is that the visual system undercomes a long period of maturation, during which can be significantly impacted by sensory experience. This post-natal visual experience is fundamental to define the cortical architecture of the visual system. Looking at it from a different perspective, we have a gradient or part of the visual system that is already there few weeks after conception, something intrinsically determined. Indeed, the descending pathway matures with sensory experience. The brain can adapt to the environment, but there is the other ascending component that is intrinsically present in the brain, and this is something you can't change. Scritto un paio di volte perchè dice sia importante.

The retinotopic organization is actually spreading its organization along the ventral and dorsal pathway, so this kind of topography and spatial embedded information is maintaining the organization of the whole brain. Progressively in the human brain we had been starting to define and parcel all the anatomically functional portions of the visual cortex (V1, V2, V3, V4, V5). This is actually derived from studies on the monkey brain, that is probably one of the best, having all the different portions being something different.

We have been describing about this for two reasons, one is that most of we know of the visual system has been studied first in monkeys, and then is the fact that we studied lesions to different regions, seeing the result. For many years we have been trying to describe this hierarchical processing of information, looking at dimension, and the idea that we always got is that we have a hierarchical organization of the visual system, whose information was processed from early visual areas more ventrally or dorsally into something that is going to integrate into a more complex representation. This thing is still valid.

In a study, Paolo created visual kind of stimuli that has been filtered, so no more playing with the models but with a filtered stimulus, seeing how these are actually correlated. You are creating an image of what each portion of the brain is actually viewing, and what you can do is to see how different an image is processed across different regions. V4 for example ignores the background information. Nowadays what you can do is to ask artificial intelligence to reconstruct the best representation that mirrors the specific response in the brain area.



As for the two pathways, these are their different properties:

Core characteristics of the *perception–action* model

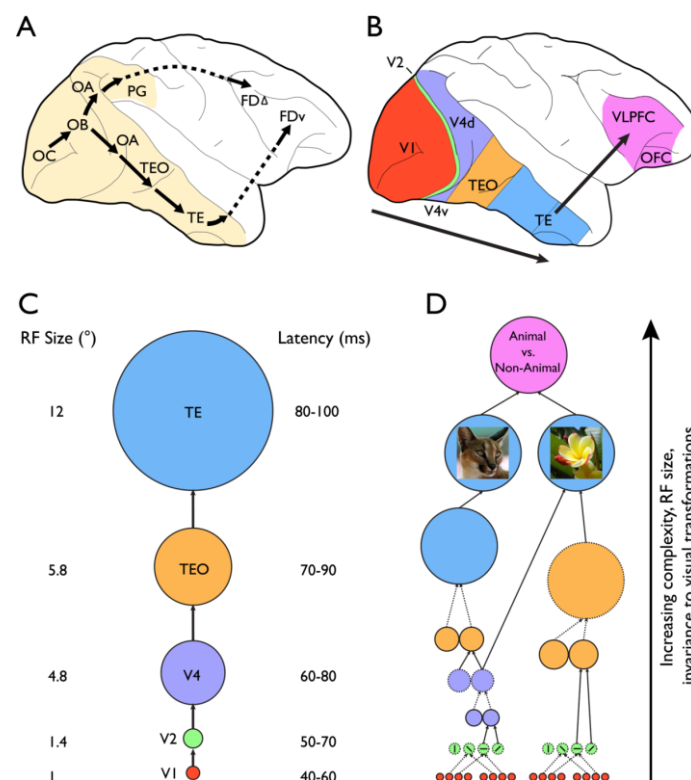
	<i>Ventral stream</i>	<i>Dorsal stream</i>
Behavioral function	Vision for perception	Vision for action
Spatial properties	Allocentric coding/ relative metrics	Egocentric coding/ absolute metrics
Temporal properties	Sustained representations	Transient representations
Visual awareness	Critically linked to awareness	Independent of awareness

Allocentric means that you are looking from outside.

Being linked to awareness means that if you see a bottle, for example, you are able to label it as a bottle.

The dorsal stream has a more egocentric coding, because it's processing action that can be performed on the object, but this is much more independent from awareness. When you are viewing something, you are not actually aware of the action you can perform on it.

If you look at the ventral stream it appears clear that the receptive field size is increasing. It's something that we had already observed for the somatosensory, at the beginning the perception is related to few receptors and the more I gather information from more receptors, I'm integrating information into a broader unique percept, and for vision it's the same. V1 has a very small receptive field size, and if I move more anteriorly the receptive field size is increasing.



For many years we have been debating how the cortex is actually organized, and in order to better understand this, let's focus on a specific item and piece of information so that it's clear.

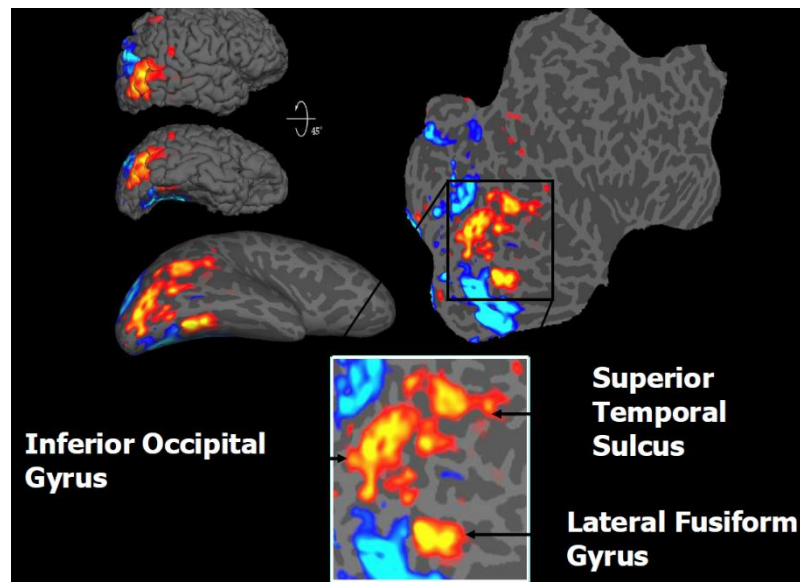
There is a very strong bottom-up component that allows us to capture what is salient in our surrounding environment.

Faces are very special, because there are plenty pieces of evidence that they are telling that one of the early stimuli, that even newborns are able to capture, is the configuration of a face. From many features of the face and face expressions, we are able to change our behaviour and thoughts about a person.

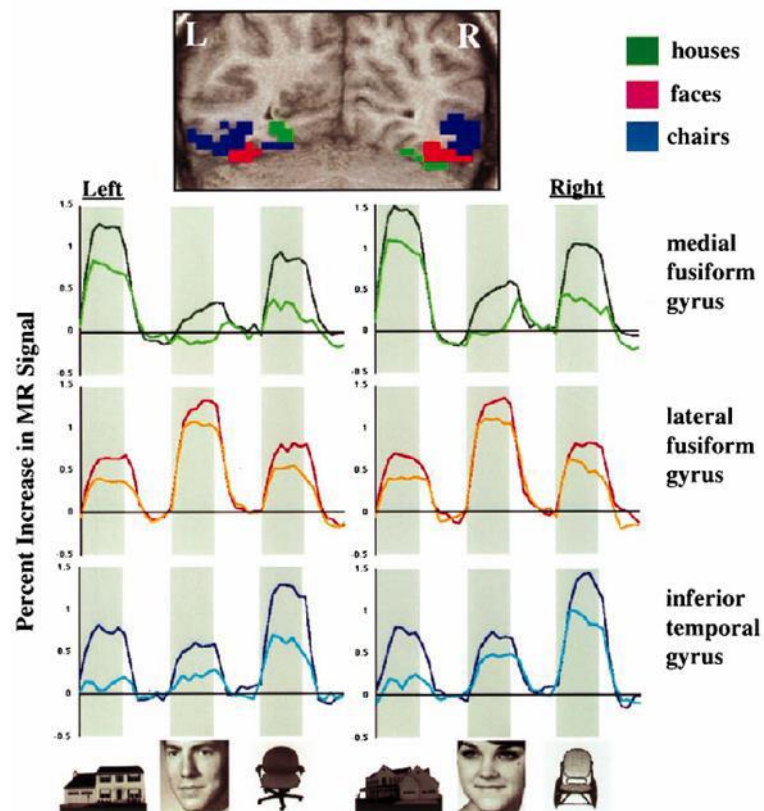
There are features of the face that are stable, and some that are changeable, and there are several additional pieces of information that allows us to say which film an actor worked for. From a face I can extract many autobiographical information.

With a study conducted on macaques it was seen that there are selected neurons that activate when seeing a face.

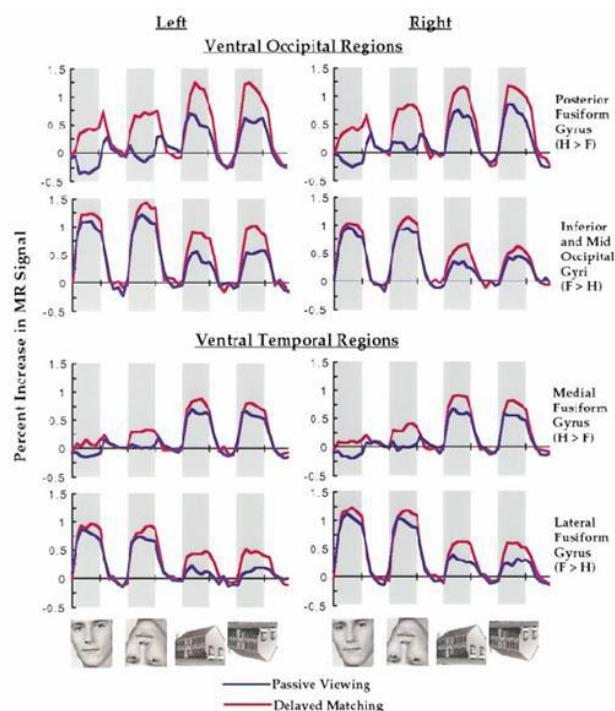
In humans in the fusiform gyrus there is an area that responds more to faces, and this is called the FFA (fusiform face area).



We are also able to detect which part of the brain responds the most to different objects, since there are neurons that activate in certain regions when we are in presence of it.



There are three main groups that create a response, faces, houses and human made objects. The darker line is when you are doing a task in which you are doing perception + attention memory, or passive viewing and no attention (no memory), that represents the lighter line. The response to houses is stronger in the medial fusiform gyrus, para hippocampal gyrus, the lateral fusiform gyrus appears to respond more for faces, and finally the inferior temporal gyrus appears to respond the most for objects.



This made evident that in few years we were able to classify various models in the ventral stream of regions dedicated to the classifying of different things.

The important question is: face is biological relevant, house and body parts also, but how many biological relevant categories can our brain fit? The cortex is not large enough to fit all different categories.

There were researchers that spent decades trying to find the face area. In different studies they tried to compare the responds in brain to face with respect to animals and objects, or living vs. non-living, animate vs. non-animate, it was gradually carried to higher representation.

Studies found that if you have subjects that are birds and cars expert, and you try to make them do a comparison between the two things, the fusiform will respond in car expert when they see a care, while in bird expert they would have a fusiform response only for birds and not cars. But this is not a good stimulus, since they both have a “face”.

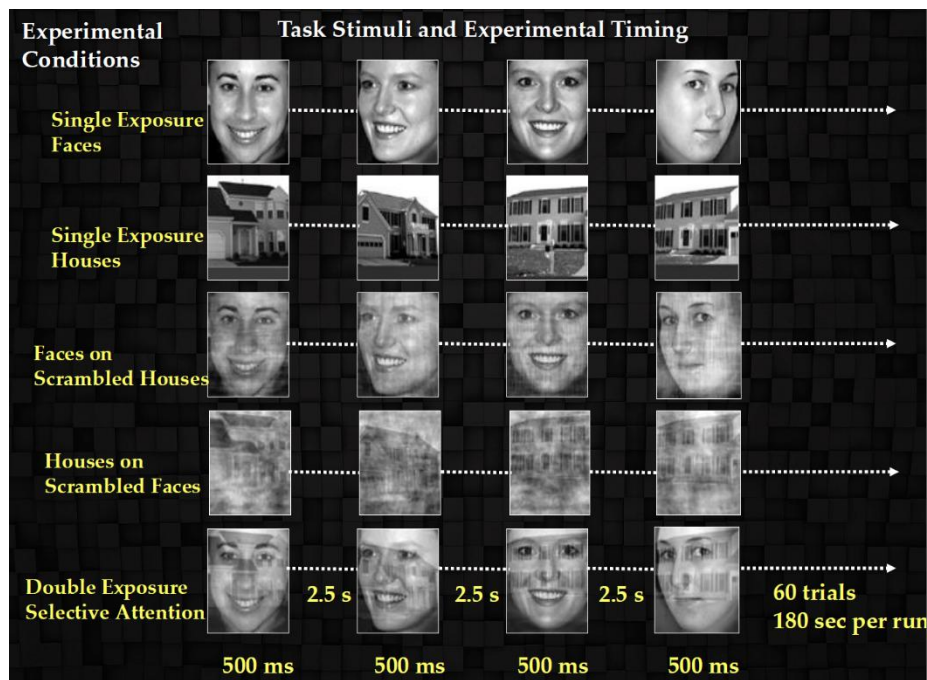
Our brain tries to predict specific shape we are familiar with.

In a study researchers grew some monkeys with the mask on, so that monkeys wouldn't have never seen a face in their life. Consequently, face deprived monkeys were not actually showing specific brain areas or response to the face. They had response for limbs but not face.

Experience is fundamental for this expertise to develop, but if I have the possibility then to study and restore face perception in these monkeys or in a baby, where would be located face perception?

In a study with babies, they were presented different stimuli, and when the baby started responding to faces it was located in the fusiform gyrus, and the same for the PPA. It means that the brain has a proto-organization, architecture, that already scaffolds functionally this ventral stream, and each part of the ventral stream appears to be innately able to compute specific information, for instance face processing. But post-natal sensory experience is necessary for this functional response to develop. If I'm exposed to a face, face processing would occur only in the fusiform face area, in the case I am exposed to places, places representation will be located in the parieto-campal place area, and not in other portions of the brain. That is way we say it is proto, it is genetically determined in the brain.

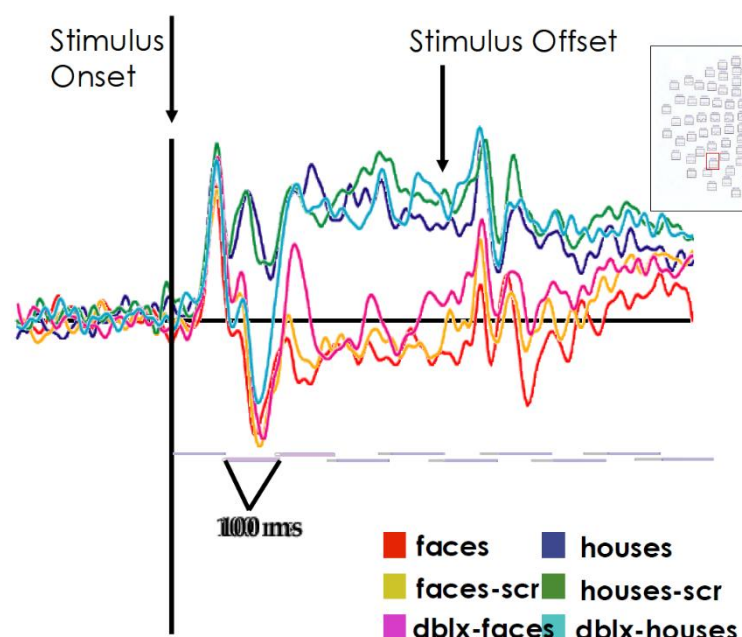
Event-related potential is an average signal that you get from the brain when you are presented several times with the same stimulus, and you record the electrophysiological response, and then you average all of them. For each potential you search for negative and positive peaks, and they called N20 because it is a negative peak after 20 milliseconds (also N17).



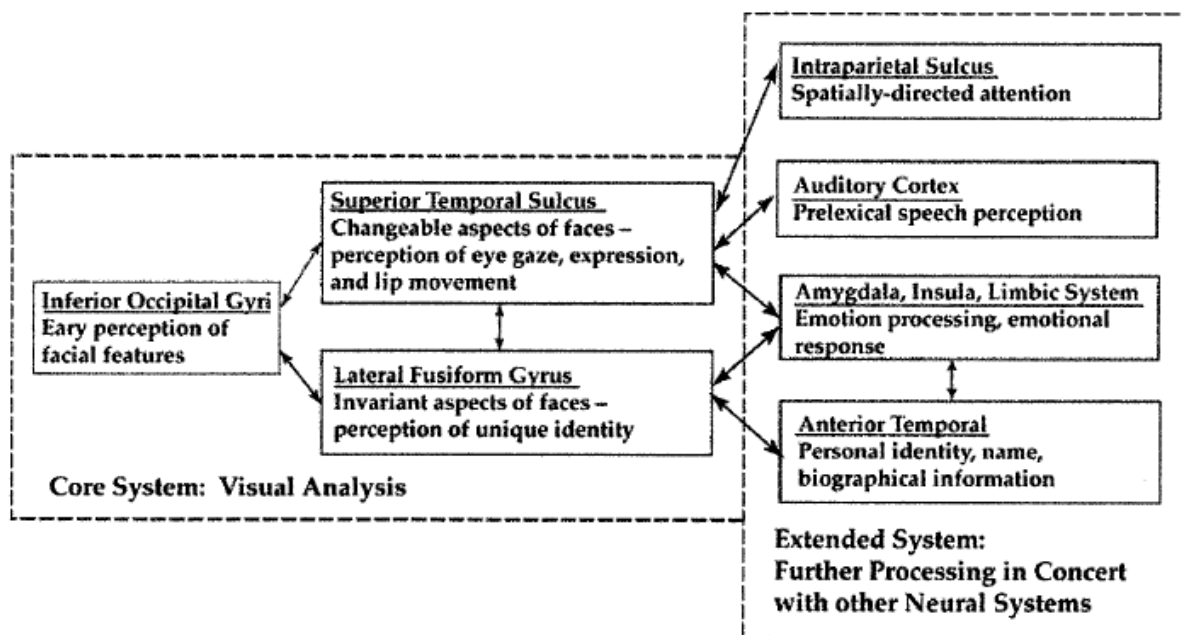
You can have a single exposure to faces or you can ask the patient to say if a face is identical or different from a previous one. And then you can have a check.

The one difficult to see, say to us one thing. When we look at those pictures you do the same task looking at faces or houses, but looking at the house is much more difficult.

The red one is for faces; with a single stimulus you have a negative peak at 100ms. Then you have the blue one for houses, with a positive peak very early that is related to the visual stimulus, and if you look at the temporal occipital cortex you see two pathways, one that is clearly faces and one that is clearly houses. The most interesting part is when it is shown the double exposure figure, asking to perform a recognition of the house, since immediately you have a very bottom-up processing, so at the beginning your brain is captured by the most salient stimulus and it starts processing the face. But then, your top-down modulation enters, because you know that the task is for house.



Getting all together it's clear that we have brain areas that are actually capturing the early features of the face. Then we have the lateral fusiform gyrus that appears to capture the invariant aspects of the brain, like identity, and then we have portion like the superior temporal sulcus, that gets all the changeable aspects of the face, like gaze, eye movements. Finally, we might have other regions that may be involved in integrating further information about face, for instance emotional processing. For face perception we have been defining what is called a core system, that is extracting the basic features of a face, and what is called an extended system (superior temporal sulcus, for example).



Different regions appear to respond more to one category than others, and even if you lower the threshold the response would be different from zero, but you have to see the peak of response to actually know what is exactly responding to. The response is much more distributed broader.

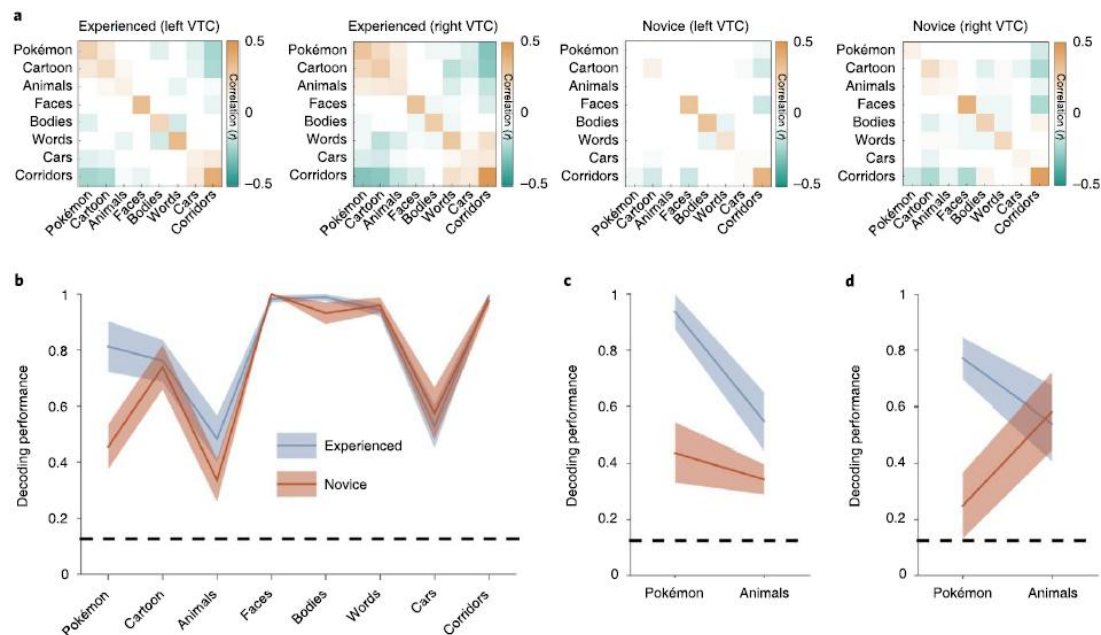
But how can the brain actually represent thousands of different categories of objects?

One issue is related to the importance of where the stimulus is represented. If you look at the ventral pathway, we found several overlapped elements, for instance faces are represented in the lateral fusiform, and places in the medial fusiform of the parieto-campal, but the same kind of discrimination was also possible if we compare animals to tools, and broadly if I contrast animate with inanimate my domain specificity is lost and we are gathering much more an imaging. If you look at objects size, typically faces are smaller, so my discrimination can be real world objects size.

Nonetheless one of the first elements that is important to represent different stimuli, it's the eccentricity bias, that is the idea that the topography organization of retinotopy extends also towards the ventral stream. So, elements that are much more foveal are represented in the lateral fusiform, and elements that are more peripheral are represented more medial. This bias of the retina gets also to the ventral stream. This is why representations overlap, and they probably have something in common; faces are typically represented more foveal (we try to focus on a face), while a place is much more relying on the periphery.

There is a study, relying on Pokémon representation (Pokémon are well distinguished between each other). Experienced people were actually very good in representing Pokémon, while novices

were not good representing them in the brain (also other categories, animals, faces). If we look at the regular image of Pokémon, they are still foveal, and corridors and space had a larger peripheral representation. So, looking at the cortical map, the Pokémon pictures should be able to be more foveal than faces, represented even more laterally.



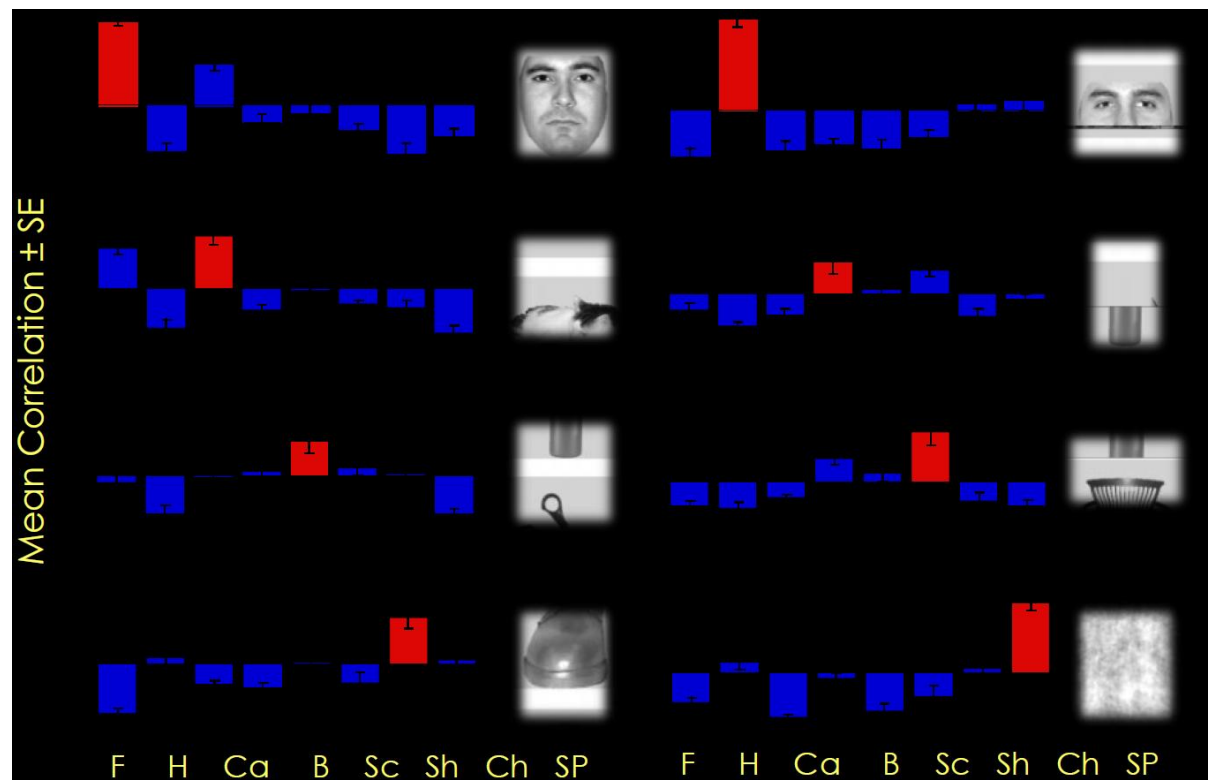
If you are looking at some other elements, like rectilinearity (the amount of linearity of representation), Pokémon should be represented even more medially than face and corridors.

They had to test models of representation of novel categories, so that we are able to know which is the feature that drives my representation. When we look specifically in experts where Pokémon are represented, it seems like that they even represent them more laterally than face, so probably the way also categories are represented and processed at the level of the brain is impacted by the way the eye and the retina (parvocellular pathway) is conveying the information to the ventral stream. This certified that we could accommodate novel categories.

But how can we accommodate thousands of categories? The idea of a researcher was to merely look at responses on a broader chunk of the cortex, look how the whole ventral-temporal cortex was responding to different categories. He used something similar to machine-learning, he divided the response that he recorded in two samples and was using one as a training and the other as a test. He was finally able to describe a specific pattern of response for each category, and then he created an average across categories and then subtracted the average from each category specific response to normalize the response and better highlight the areas with a higher or lower response to the mean. Therefore, you can recognize that for face, the higher response is in the fusiform gyrus, and that for places and houses the higher response is in the medial fusiform parahippocampal gyrus, something that we are already aware of. If you specifically look at correlation between different responses, seems like two responses are correlating higher, so something is likely to be a face. And he did this work for all different categories.

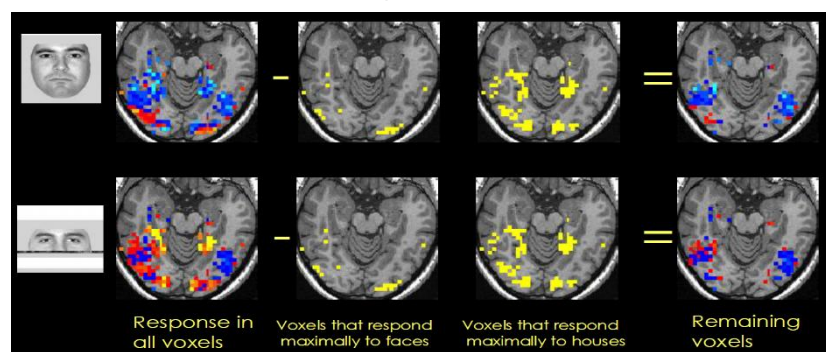
He showed that for a face the higher correlation is for the face, and less for all other categories, and what is interesting is that there is also a slide correlation with animated cats. He specifically defined category specific responses, not as a single module and regions, but as a pattern of response. He didn't limit to a specific representation but he went for the whole pattern, so the

difference was moving from a module to a map, from an area to a pattern of response. It looked like that the pattern was much more able to capture elements of a broader representation of category, and each pattern was specific for a domain for a category.



Criticisms?

These patterns are all distributed, and they are also overlapping. If I want to correlate and to show that a pattern is actually unique for an element, what can drive the correlation? Probably the peak response. Likely, the peak response of the fusiform gyrus could drive the whole correlation, and therefore if this information is really distributed across the whole pattern, so the whole pattern is significant for recognizing a face, I should be able to remove the peak response and get the discrimination across categories. Consequently, what the researcher did was to take each peak response (voxel that responds maximally for the element) and remove them. The remain voxels should be equally able to discriminate each category, if the information is truly distributed, and even if the accuracy of classification is reducing, you are still able to get higher accuracy across categories. If the information is truly distributed, then, it should do also the opposite and limit the classification and discrimination within the maximal responding voxels and be able to classify the other categories. So, you could go in the fusiform face area and rely on those voxels to classify also houses, bottles, cars. And this is actually what we are able to do.



So, the novel concept (in the study called object form topography) is that the information that is present in the ventral stream is probably more distributed than we expected, and also a little bit related to the kind of approach that we adopt. The idea is indeed not focusing anymore on few voxels that respond maximally to a piece of information, to find the category selective regions, but to go to a much more distributed representation of a domain of category. Therefore, I'm no more focusing on the fusiform face area, but I'm actually thinking that a face is represented broadly across the cortex, and this is a huge pro, because the number of maps and patterns that I'm going to have is illimited, and I don't have to concern anymore with which feature are relevant in representing the cortex, because all elements could be represented here and the whole pro is that patterns of categories that are semantically similar (faces and cats) should be somehow similar and a little bit overlapping in some of the features, but still distinguishable between them.

An aspect less clear is, why for many years we had been able to describe disorders that are very selective? If we have a face selective area lesioned, we have prosopon agnosia, and the information is distributed all across the cortex, why a lesion of the fusiform face area is producing the inability to recognize a face. We can claim several things, but probably we are really looking at the same response with two different kinds of analytical tools, highlighting different perspectives of my brain organization. Probably there is something that is in the middle.

The main difference, in what we talked about, is that if we look at the single voxel, you have a single musical instrument playing, if you look at the pattern you are listening the whole orchestra playing. And finally, what this study is telling us is that for sure the information about a specific category is distributed and involves more regions, but probably some regions are contributing more to the representation than others.

In order to merge this kind of view, we should actually think that the ventral stream may continue a sort of like hierarchical and nested information, and just a matter of were I'm actually focusing. So, probably you have in the brain different areas related to the response for single elements, and all this elements are responding to something animate, but this pattern of response is actually integrating different pieces of information, and the more you broaden your map the more is the possibility to represent diversity, but at the same time the risk is to miss the specificity of the response that our brain is able to. Furthermore, linguistically I'm able to organize these domains and categories at different level of complexity and comprehension, and likely this way of organizing knowledge that we express in language is also the same way the brain is organizing knowledge within the cortex, defining a sort of hierarchical and nested representation of different elements, that then gather together in a broader super-ordinate element.

Obviously methodologically I could drive my interpretation; if you decide to look at the connectivity between regions, the theoretical model that you have is that information is connecting different brain areas and pass one area from another. If you look at one small portion of the brain and you ignore the rest, obviously your idea is that that piece of brain contains meaningful information for your task.

There is not a good or a bad approach, if you decide to find a threshold and get the higher response of your voxels, your idea is much more the one of looking at the modules doing different things. If instead, you lower your threshold and you look at broader number of voxels distributed in a wide extent of the cortex you are looking at the map, the distributed representation. The first one is very good for any univariate approach, and the second one for any multivariate approach, from a statistical perspective.

One thing that you should always keep in mind, is how much your methodological approach is impacting on your theoretical model. Or you can have a strong theoretical model that will guide in using on methodological approach or the other. But beware that once you look at the results you should keep in mind that they come from a specific methodological approach, and this could drive the interpretation of data.

There may be features that are representing different elements and you can actually use this according to the different categories (face for humans but also for cats, body is not). It's probably that we have simple different elements that are summing up to represent different elements.

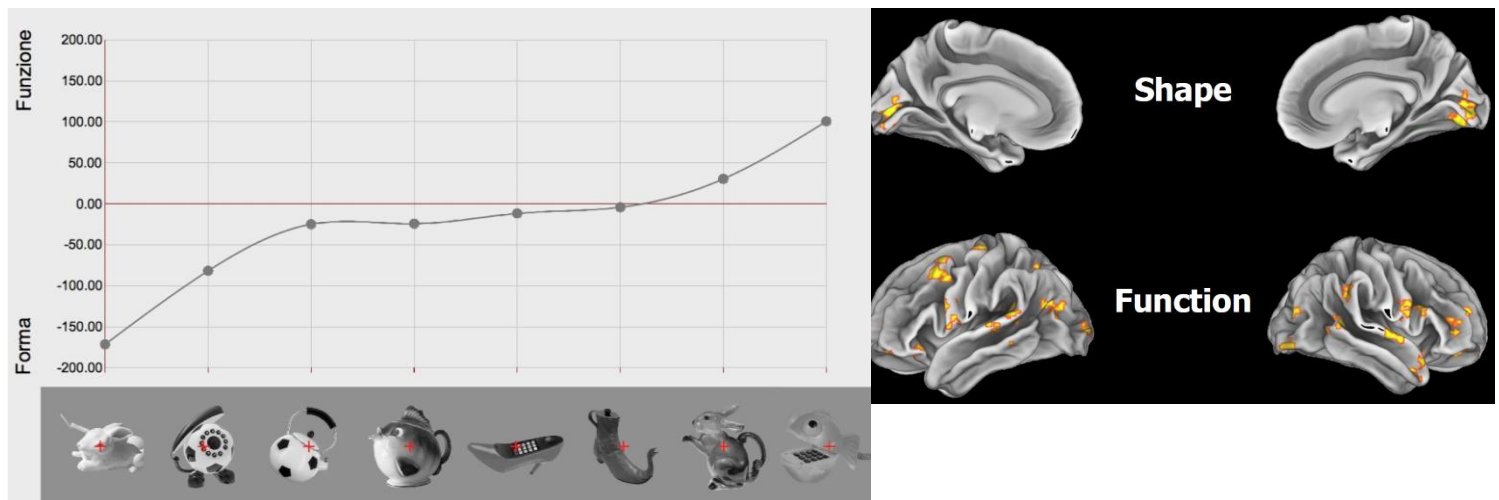
The deep network has different layers that extract different piece of information from the stimulus you are processing, and these layers often overlap with the hierarchical processing of vision, and so V1 is much closer to early layers in our network, while advanced regions are going much deeper, so there are several attempts to module on this way. Other approaches are relying on language, using it as a model to test the complexity of the representation. So, nowadays computation tends to work a lot, we should think about doing the opposite thing. If my representation is not primarily done by different features, but actually what I am going to represent is much more related, for instance, to action and verbs that are much more related to my stimuli, this is much more related to encoding approaches and are equally valid to our purpose.

If we have different verbs (bark, fly, run, eat) and then you see a dog, it will be in different ways correlated with these different verbs (bark 0.6, fly 0.01, run 0.4). If we have a specific voxel that are processing the verbs, just with giving a weight to these voxels you can represent a dog. For an apple we can do the same thing. This kind of approach (encoding of the information) is working on the opposite way, so you define the criteria organization of your ventral cortex and you just weight the representation on the cortex per se, and this kind of thing is impressive, because you can find and replicate the verbs or elements semantically related to house, for example, would be represented much more on the parahippocampal area. This is actually how you can process the information. Is the brain actually operating in terms of these 25-20 verbs? No, but this is how you define and decide to describe semantically this organization of knowledge and is actually impacting you theoretical model of the brain, that is much more related to the events that you are doing with these objects, other than on their shape, texture or colour.



This is not a pipe because you can't use it to smoke, and you can create hybrid objects like this. When you try to understand and describe this element, this is something "pipe-shape class", and the idea is that you try to disentangle the shape and the function. With this type of objects, you can define a behavioural curve, that is telling you whether each object is much more driven by the shape or the function. When you try to look at the response and weight according to these elements, when you see that the shape is something that you can detect easily in the early visual areas and a little bit in the ventral stream, but function is indeed extended more broadly. For a

stimulus, if you look at which features are represented according to the behavioural description, you see that information about shape is just limited to the ventral stream, but when you talk about something more related about function and action, there is a broader engagement of the whole brain.



The third pathway is much more related to the superior temporal sulcus involved in social perception (recognizing facial expression, aspects useful for social interaction), a theory of just 1-2 papers. Obviously, all the information of the two pathways is highly integrated constantly, to get more complex and high-level representation.

Supramodality

Neuroscience started to consider other sensory modalities. Do we really need vision to capture all different pieces of information? When we listen to sounds that we know, our ventral stream is processing even auditorily the sound, but without being aware our brain is also processing the action that produced the sound, we do not truly need vision to make sense of the surrounding world. It seems like that several features are not actually depending on visions, sight has always been regarded as the most important sensory modality, and more than 50% of the brain is devoted to process vision, nonetheless all those who are visual deprived since birth appear to show most of their perceptual and social skills similar to those found in other individuals. There are no superior skills, they simply rely on other sensory modalities and some kind of external world representation are almost similar with some difference.

It's important to mention the spatial alignment task, which is something that can be done when you create a sort of discrepancy in congruency between the vision and motor command. You can have a conditional incongruency, so the command is to perform the action with the right hand, but the ball is on the left, this creates a small delay in the response. The ball can also be in a reachable distance or not reachable, if the ball is in a space that is not reachable this effect disappears. When the object is outside our space of action, we are not processing it anymore.

The first attempt was to translate the task in an auditory one, so people were blindfolded and the ball was bouncing on the left or the right. Some individuals could present the same effect on the auditory condition, but also congenitally blind people. So blind people that never experienced vision have a sort of representation of the space around them.